

MID Semester of 3rd Sem B.Tech.

MID –SEMESTER EXAMINATION, SEPTEMBER, 2025

COURSE CODE: EAMTC04 (Backlog)

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 25

Note:- Attempt all questions. Missing data/ Information (if any), may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	$T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - y, 2z)$. Check if T is linear transformation.	2.5	CO1
1b	Examine if the set $W = \{\alpha(1,1,1): \alpha \in R\}$ is a subspace of R^3 .	2.5	CO1
2a	Find the matrix of the linear transformation T on R^3 defined as $T(x, y, z) = (2y + z, x - 4y, 3x)$ with respect to the ordered basis $\{(1,1,1), (1,1,0), (1,0,0)\}$	2.5	CO1
2b	Find $\text{Ker}T$ and $\dim(\text{ker}T)$ of the linear mapping $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x, y)$.	2.5	CO1
3a	Let $W = \{(1,3), (-2,6)\}$. Is W basis of R^2 . Justify your answer.	2.5	CO1
3b	Show that $S = \{(2,-1,3), (-1,1,1), (-4,-5,1)\}$ is orthogonal basis of R^3 . Find the orthonormal basis of R^3 .	2.5	CO2
4a	Check the nature the quadratic form $2x^2 + 3y^2 + 2xy + 2z^2$.	2.5	CO2
4b	Find Eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$.	2.5	CO2
5a	Use Gram-Schmidt process to convert the basis $\{(1,2,3), (1,1,1), (1,0,1)\}$ of the Euclidean space R^3 with standard inner product into an orthogonal basis.	2.5	CO2
5b	If 2, 3 are the eigen values of $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ a & 0 & 2 \end{bmatrix}$, find the value of a.	2.5	CO2

	requires 24.60 mL of EDTA solution to reach the end point (using Eriochrome Black T as indicator). Next, 50.00 mL of a tap water sample is titrated with the same standardized EDTA solution and requires 18.75 mL of EDTA to reach the end point. Calculate the hardness of the given water sample in mg L^{-1} as CaCO_3 equivalents.		
2b	100 mL of water sample on titration with $\text{N}/50$ HCl required 5.0 mL of the acid up to the phenolphthalein end point. A few drops of methyl orange were added to the same solution, and the titration was further continued. The yellow colour of the solution just turned red after the addition of another 11.0 mL of the acid solution. Calculate the ions responsible and the extent of total alkalinity and alkalinity due to individual ions in ppm.	5	2
2c	i) State any two advantages of ion exchange process over zeolite process used for removal of permanent hardness of water. ii) Write the reaction involved in regeneration of exhausted zeolite. iii) Convert 300 ppm hardness into degree Clark and degree French units.	2+1+2	2
Q 3	Attempt any 2 parts of the following		
3a	(a) Justify with reaction(s) that unnecessary derivatization should be minimized or avoided if possible. (b) Which principle discusses "Design for Degradation"? Give one example that violates the same.	3+1+1	3
3b	The maleic anhydride is synthesised using two methods Method I: Benzene Oxidation  Method II: Butene Oxidation  Calculate the atom economy of the reactions. Which method is greener? Give two reasons in support of your answer.	4+1	3
3c	i Who is the father of Green Chemistry? ii State the greener way to get clear water from turbid water? iii Which drug causes birth defects (including missing or deformed limbs) that opposes principle no. 4? iv Which is the widely used natural insecticide extracted from the chrysanthemum flower? v Name the green plastic that is made from corn?	5	3
Q 4	Attempt any 2 parts of the following		

4a	(i) Explain the construction and working principle of a hydrogen fuel cell with a neat diagram. (ii) Describe the synthesis of polylactic acid (PLA) through lactide formation.	2+3	4
4b	(i) Explain the process of Photocatalytic water splitting using chemical reactions. (ii) Give any two important applications of PVA. (iii) Explain why PVA cannot be synthesized directly from its corresponding monomer.	3+1+1	4
4c	Draw a flow chart to explain how biodiesel is obtained from Jatropa oil? Also, write the chemical reaction involved.	3+2	4
Q 5	Attempt any 2 parts of the following		
5a	(i) The absorption coefficient of a glycogen-iodine complex is $0.20 \text{ mol}^{-1} \text{ L cm}^{-1}$ at a wavelength of 450 nm. What is the concentration when the transmission is 40% in a cuvette of thickness 2 cm? (ii) Mention all the possible UV-Visible electronic excitations that can occur in a nitroethane molecule. (iii) How does the particle size of the adsorbent in column packing affect the column efficiency? Also mention the usual particle size range.	2+1+2	5
5b	(i) A solution of chemical A having concentration 0.18 mol L^{-1} , has an absorbance of 0.48. Another (second) solution of A under the same conditions has an absorbance of 0.27. What is the concentration of second solution of A? (ii) In a paper chromatography experiment, the solvent front moved 15.0 cm from the origin. Two solute spots, X and Y, were found at distances of 9.0 cm and 4.5 cm from the origin, respectively. Answer the following: (a) Calculate the R_f values of X and Y. (b) State which solute is more polar, and give reason.	2+ 2+1	5
5c	(i) Mention two examples of commonly used adsorbents for adsorption chromatography. State which will elute move slowly from column among ethers and amines? (ii) What is the main difference between hyperchromic shift and hypochromic shift? (iii) A shift to higher energy is..... while a shift to lower energy is.....	3+2	5